

Problems

- Q1.** Construct a 3×3 unitary matrix U such that the first column is proportional to $(i, i, \sqrt{2})^T$. Show that $U^*U = I$.
- Q2.** Let $V = \mathbb{R}^2$ and define a bilinear form B by $B(x, y) = 2x_1y_1 + 3x_1y_2 + 3x_2y_1 + 4x_2y_2$.
- (a) Find the matrix A representing B with respect to the standard basis.
 - (b) Find the matrix A' representing B with respect to the basis $\mathcal{B} = \{(1, 1), (1, -1)\}$.
 - (c) Verify the relation $A' = P^TAP$ for the appropriate transition matrix P .
- Q3.** Let $A = (a_{ij})$ be a real symmetric positive definite matrix. Prove that its maximal entries are on the diagonal.
- (Hint:**
- (a) By considering the vector $x = e_i$, show that all diagonal entries a_{ii} are positive.
 - (b) By considering the principal 2×2 submatrices (or by examining the quadratic form $x^T Ax$ for vectors with only two non-zero components), prove that $a_{ij}^2 < a_{ii}a_{jj}$ for all $i \neq j$.
 - (c) Conclude that the entry with the largest absolute value in A must be on the diagonal.)
- Q4.** Show that any bilinear form B on a vector space V can be uniquely written as $B = B_s + B_a$, where B_s is symmetric and B_a is skew-symmetric. Provide explicit formulas for $B_s(v, w)$ and $B_a(v, w)$.
- Q5.** Prove that if A is an invertible matrix then A^*A is Hermitian and positive definite.
- Q6.** Show that if B is a non-degenerate skew-symmetric form on V , then $\dim(V)$ must be even.
- Q7.** Show that the product of two orthogonal projections is an orthogonal projection if and only if the projections commute.
- Q8.** Find an orthonormal basis for the vector space P of all real polynomials of degree at most 2, with the symmetric form defined by

$$\int_{-1}^1 f(x)g(x)dx.$$

- Q9.** Let A, B be complex $n \times n$ matrices.
- (a) Decide whether or not the rule $(A, B) = \text{trace}(A^*B)$ defines a Hermitian form on the space $\mathbb{C}^{n \times n}$ of complex matrices, and if so, determine its signature.
 - (b) Answer the same question for the form defined by $(A, B) = \text{trace}(\overline{A}B)$.

Q10. Let $w \in \mathbb{R}^n$ be a vector of length 1, and let U denote the orthogonal space $w^\perp$. The reflection r_w about U is defined as follows: We write a vector v in the form $v = cw + u$, where $u \in U$. Then $r_w(v) = -cw + u$.

- (a) Prove that the matrix $P = I - 2ww^t$ is orthogonal.
- (b) Prove that multiplication by P is a reflection about the orthogonal space U .
- (c) Let u, v be vectors of equal length in $\mathbb{R}^n$. Determine a vector w such that $Pu = v$.